aircraft-engine system drag and, consequently, increased aircraft fuel burn. The overall impact of increasing bypass ratio is a system optimum that is at a much lower bypass ratio than one finds when only considering the engine.

7.5 Turbofan with Afterburning: Mixed Exhaust Stream

The dual-spool, mixed-flow, augmented, low-bypass turbofan is the engine cycle of choice for all modern high-performance fighter aircraft and supersonic bombers like the B-1B. Figures 1.9a and 1.9b show cross-sectional drawings of this engine type in use today. (Refer to Appendix B for examples of the bypass ratios used in these engines.) Most high-speed military aircraft require an afterburner for thrust augmentation at certain flight conditions (e.g., takeoff, high-speed turns, etc.). Figure 7.20a shows the cross section of a mixed-flow afterburning turbofan engine, and Fig. 7.20b shows a T-s plot for this cycle. The mixed-flow turbofan cycle has several advantages over the turbofan with separate exhausts: 1) one variable-area exhaust nozzle, 2) one augmenter (afterburner), and 3) cold bypass air available to cool the afterburner liner and nozzle, and/or for low observables (LO) considerations. Also, mixers can improve engine performance when the afterburner is off or not present. All of these advantages are, of course, at the expense of added complexity in terms of both analysis of the cycle performance and the design and construction of the engine to realize that performance.

7.5.1 Mixer

In our modeling of the ideal cycle for the mixed-flow afterburning turbofan engine (see Chapter 5), we first defined the temperature and pressure ratios of the mixer as

$$\tau_M = \frac{T_{t6A}}{T_{t6}} \qquad \pi_M = \frac{P_{t6A}}{P_{t6}} \tag{5.49}$$

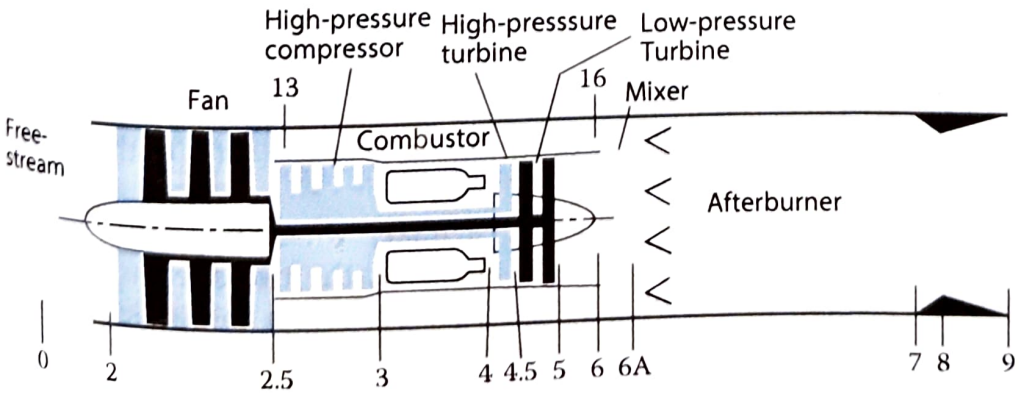


Fig. 7.20a Station numbering for mixed-flow turbofan engine with afterburner.

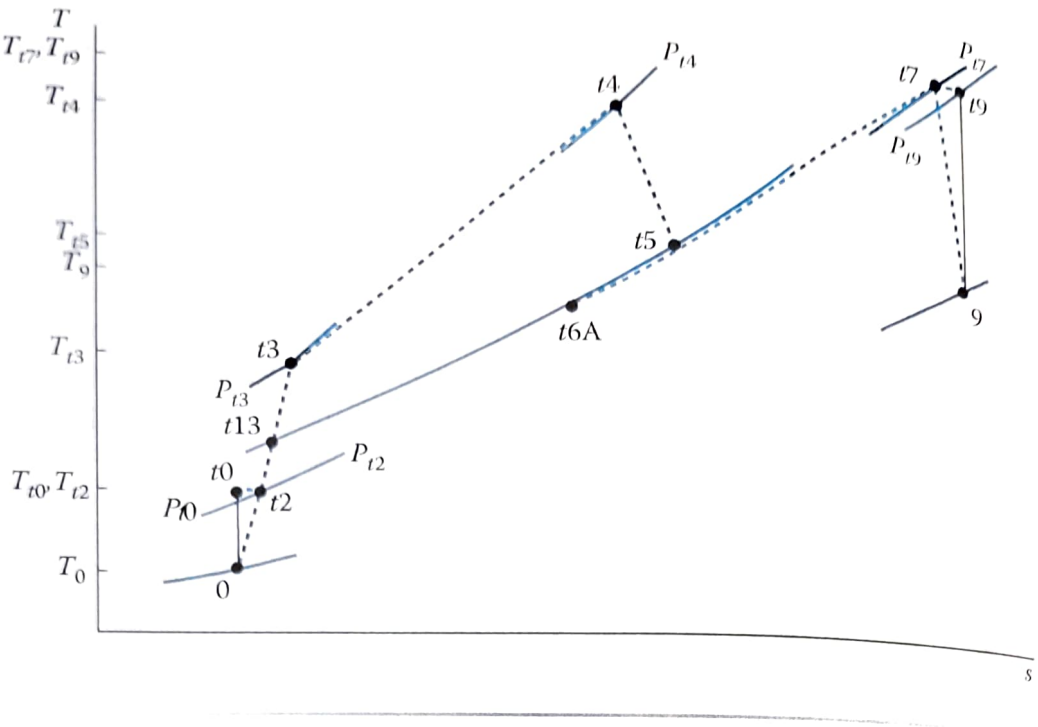


Fig. 7.20b *T-s* diagram for mixed-flow turbofan engine with afterburner.

We also assumed that the total pressures of the two entering streams are equal ($P_{t6} = P_{t16}$) and that the mixer pressure ratio π_M was unity. The mixer temperature ratio τ_M for the ideal cycle was obtained from an energy balance of the mixer.

The mixer temperature and pressure ratios defined in Eq. (5.49) will be used in the analysis of this engine cycle with losses, using the analysis presented in Section 6.9. Our mixer model is shown in Fig. 7.21, a constant-area mixer. The mixer temperature ratio τ_M was obtained from an energy balance of the mixer. In addition, we assume that the mixer pressure ratio π_M is the product of the pressure ratio of an ideal constant-area mixer ($\pi_{M\,\text{ideal}}$) times a total pressure ratio for the frictional losses ($\pi_{M\,\max}$)

$$\pi_M = \pi_{M\,\max}\,\pi_{M\,\text{ideal}} \tag{7.60}$$

The analysis in Section 6.9 obtained the total pressure ratio of the ideal mixer $\pi_{M\text{ideal}}$ [Eq. (6.38)] and the total temperature ratio of the mixer τ_M [Eq. (6.31)].

Figure 7.21 shows a constant-area mixer where the core and bypass streams are mixed together. At the entrance, the two streams are divided by a splitter plate that suddenly ends. Nature forces the static pressure of the two streams to be equal (Kutta condition)—just like the trailing edge of a wing where the two streams come together. Thus we have

$$P_{16} = P_6 \tag{7.61}$$

The total pressure of each stream could be independent variables but the best performance of the mixed-flow turbofan engine occurs when the two total pressures are nearly equal (Refs. 13 and 23). Thus, for ease of analysis, we will consider that

$$P_{t16} = P_{t6} \tag{7.62}$$

As a result of this selection, the fan pressure ratio π_f and bypass ratio α are no longer independent variables. We get to select one, and the other one is calculated so that Eq. (7.62) is satisfied.

7.5.2 Cycle Analysis

This secton develops the additional equations needed to analyze the mixed-flow afterburning turbofan engine. Because this engine, like the turbojet, has a single inlet and exhaust, Eq. (7.1) can be used for the engine uninstalled thrust, and Eq. (7.2) relates the pressure unbalance term to other flow quantities. We define the overall fuel/air ratio for this engine as the total fuel flow rate (main burner plus afterburner) divided by the inlet airflow rate, or

$$f_O = \frac{\dot{m}_f + \dot{m}_{f\text{AB}}}{\dot{m}_0} \tag{7.63}$$

We assume in this analysis that the mass flow rate at station 9 equals the mass flow rate at station 0 plus the fuel added in the main burner and in the afterburner. In equation form, we write

$$\dot{m}_9 = \dot{m}_0 + \dot{m}_f + \dot{m}_{f\text{AB}}$$

or

$$\frac{\dot{m}_9}{\dot{m}_0} = 1 + f_O \tag{7.64}$$

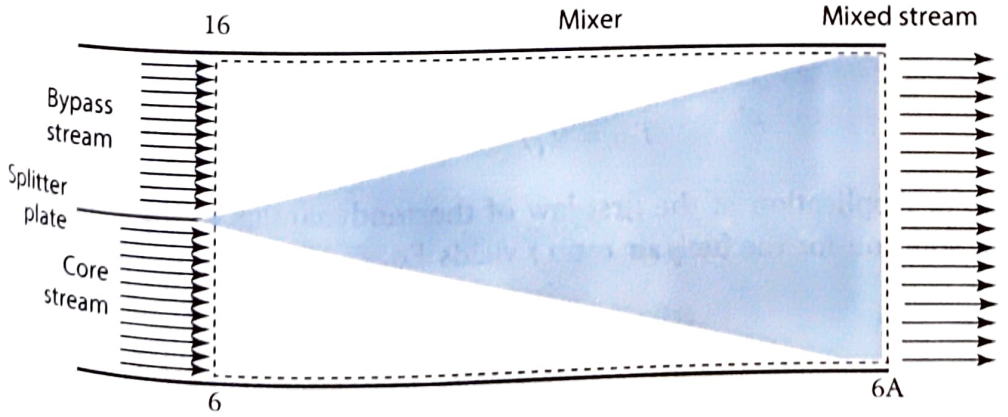


Fig. 7.21 Model of constant-area mixer.

In the following, we use the steps of cycle analysis as listed in Section 5.4.

Step 1: Uninstalled thrust: Using Eqs. (7.1) and (7.2), along with Eq. (7.64), we can write

$$\frac{F}{\dot{m}_0} = \frac{a_0}{g_c}\left[(1+f_O)\frac{V_9}{a_0} - M_0 + (1+f_O)\frac{R_9}{R_c}\frac{T_9/T_0}{V_9/a_0}\frac{1-P_0/P_9}{\gamma_c}\right] \tag{7.65}$$

where the subscript c is used for gas properties at station 0.

Step 2:

$$\left(\frac{V_9}{a_0}\right)^2 = \frac{a_9^2 M_9^2}{a_0^2} = \frac{\gamma_9 R_9 g_c T_9}{\gamma_0 R_0 g_c T_0} M_9^2$$

Using the subscript c for the gas properties at station 0, we write

$$\left(\frac{V_9}{a_0}\right)^2 = \frac{\gamma_9 R_9 T_9}{\gamma_c R_c T_0} M_9^2 \tag{7.66}$$

Step 3: We have

$$M_9^2 = \frac{2}{\gamma_9 - 1}\left[\left(\frac{P_{t9}}{P_9}\right)^{(\gamma_9-1)/\gamma_9} - 1\right] \tag{7.67a}$$

where

$$\frac{P_{t9}}{P_9} = \frac{P_0}{P_9}\pi_r \pi_d \pi_c \pi_b \pi_t \pi_M \pi_{AB} \pi_n \tag{7.67b}$$

Step 4: We have

$$\frac{T_9}{T_0} = \frac{T_{t9}/T_0}{(P_{t9}/P_9)^{(\gamma_9-1)/\gamma_9}}$$

where

$$T_{t9} = T_{t4}\tau_t \tau_M \quad \text{afterburner off} \tag{7.68a}$$

or

$$T_{t9} = T_{t7} \quad \text{afterburner on} \tag{7.68b}$$

Step 5: Application of the first law of thermodynamics to the main burner and solution for the fuel/air ratio f yields Eq. (7.10), where

$$f = \frac{\dot{m}_f}{\dot{m}_C} \tag{7.69}$$

Application of the first law of thermodynamics to the afterburner gives

$$\dot{m}_{6A} h_{t6A} + \eta_{AB}\dot{m}_{fAB} h_{PR} = \dot{m}_7 h_{t7}$$

Dividing the preceding equation by $\dot{m}_0$ gives

$$\left(1+\frac{f}{1+\alpha}\right)h_{t6A}+f_{AB}h_{AB}h_{PR}=\left(1+\frac{f}{1+\alpha}+f_{AB}\right)h_{t7}$$

Solving for the afterburner fuel/air ratio f_{AB} gives

$$f_{AB}=\left(1+\frac{f}{1+\alpha}\right)\frac{h_{t7}-h_{t6A}}{\eta_{AB}h_{PR}-h_{t7}} \tag{7.70}$$

Iteration is required to solve this equation (like the equation for the fuel/air ratio of the main burner) because h_{t7} is a function of the overall fuel/air ratio of the mixture.

Step 6: The power balance among the turbine, compressor, and fan, with a mechanical efficiency η_m of the turbine shaft, gives Eq. (7.44). This equation, solved for the turbine temperature ratio, gives

$$\tau_t=1-\frac{1}{\eta_m(1+f)}\frac{\tau_r}{\tau_\lambda}[\tau_c-1+\alpha(\tau_f-1)] \tag{7.71}$$

This expression allows solution for τ_t, from which we then obtain π_t by using Eq. (7.13) and h_t by using Eq. (7.14). Equations (7.15), (7.16), (7.46), and (7.47) are used to obtain τ_c, η_c, τ_f, and η_f, respectively, from given π_c, π_f, and polytropic efficiencies.

As presented previously in Section 7.5.1, normally, the fan pressure ratio or bypass ratio is selected for this engine cycle so that the total pressures at stations 6 and 16 are equal (Refs. 13 and 23). Assuming isentropic flow in the bypass duct from 13 to 16, we can write

$$\pi_c\pi_b\pi_t=\pi_f$$

or

$$\tau_t=\pi_t^{(\gamma_t-1)e_t/\gamma_t}=\left(\frac{\pi_f}{\pi_c\pi_b}\right)^{(\gamma_t-1)e_t/\gamma_t} \tag{7.72}$$

Equations (7.71) and (7.72) can be solved to obtain the bypass ratio α or the fan temperature ratio τ_f in terms of known quantities. Solution for the bypass ratio gives

$$\alpha=\frac{\eta_m(1+f)(\tau_\lambda/\tau_r)\{1+[\pi_f/(\pi_c\pi_b)]^{(\gamma_t-1)e_t/\gamma_t}\}-(\tau_c-1)}{\tau_f-1} \tag{7.73}$$

Step 7: See Eq. (7.65).

Step 8: The equation for the thrust specific fuel consumption is

$$S=\frac{f_O}{F/\dot{m}_0} \tag{7.74}$$

Step 9: From the definitions of propulsive and thermal efficiency, one can easily show that, for this turbofan engine,

$$\eta_P = \frac{2g_c V_0 (F/\dot{m}_0)}{a_0^2[(1+f_O)(V_9/a_0)^2 - M_0^2]} \tag{7.75a}$$

$$\eta_{Th} = \frac{a_0^2[(1+f_O)(V_9/a_0)^2 - M_0^2]}{2g_c f_O h_{PR}} \tag{7.75b}$$

Now we have all the equations needed for analysis of the mixed-flow afterburning turbofan engine. For convenience, this system of equations is listed (in the order of calculation) in the Supporting Material for easier calculation.

See Section 7.5.SM, Summary of Equations: Mixed-Flow Afterburning Turbofan Engine, in the online supporting material for related information.

Example 7.9

The afterburning and nonafterburning performance of the mixed-flow turbofans with losses are plotted in Fig. 7.22 vs fan pressure ratio π_f at two flight Mach numbers. Figure 7.23 plots the required bypass ratio α (to meet our mixer analysis constraint of equal stream total pressures) vs compressor pressure ratio π_c for the different values of fan pressure ratio π_f at flight Mach numbers of 0.9 and 2. Calculations were performed for the following input data:

$$c_{pc} = 0.240 \text{ Btu/(lbm} \cdot {}^\circ\text{R)}, \quad \gamma_c = 1.4, \quad T_0 = 390^\circ\text{R}, \quad \pi_{d\max} = 0.98,$$
$$\pi_{M\max} = 0.98, \quad e_c = 0.90, \quad e_f = 0.89, \quad e_t = 0.91, \quad \pi_n = 0.98,$$
$$c_{pt} = 0.295 \text{ Btu/(lbm} \cdot {}^\circ\text{R)}, \quad \gamma_t = 1.3, \quad \eta_b = 0.995, \quad \pi_b = 0.96, \quad T_{t4} = 3200^\circ\text{R},$$
$$c_{pAB} = 0.295 \text{ Btu/(lbm} \cdot {}^\circ\text{R)}, \quad \gamma_{AB} = 1.3, \quad \eta_{AB} = 0.95, \quad \pi_{AB} = 0.96,$$
$$T_{t7} = 3600^\circ\text{R}, \quad h_{PR} = 18{,}400 \text{ Btu/lbm}, \quad M_6 = 0.4, \quad \eta_m = 0.99, \quad \frac{P_0}{P_9} = 1,$$
$$\pi_f = 2 \rightarrow 5, \quad M_0 = 0.9, 2$$

A compressor pressure ratio π_c of 24 was chosen for the engines of Fig. 7.22 as a reasonable choice for π_c in terms of tradeoff between specific thrust and fuel consumption (see Fig. 7.12a). Figure 7.22 is a composite performance plot of fuel consumption vs specific thrust with different fan pressure ratios and Mach numbers—this is another example of a *carpet plot* presented in a different manner than the ones up to this point. This

(Continued)